1. DATASETS & LEAKAGE CONTROL

Public datasets

- **D1** CSE-CIC-IDS2018
- **D2** UNSW-NB15
- **D3** ToN-IoT
- **D4** BoT-IoT (transfer)
- **D5** Lycos2017

Flow timestamps

Used only for ordering and splitting

Leakage control

- **Drop identifiers**
  IPs, ports, DNS IDs, TTL
- **Train-only scaling**
  Median impute, log(1+x), z-score; 41 common features
- **Time blocks**
  1-hour (all); 10-min (D5 purged split)
- **Windows**
  T = 32, stride 16
- **SHA-256 deduplication**
  Remove duplicate windows
- **Integrity checks**
  Disjoint splits, class ratios, finite values

2. PAIRED ZERO-DAY PROTOCOLS

Same data, detectors, seeds

Report infeasible LOACO folds

Leakage-controlled windows

LOACO (hold-out)

Grouped random (60/20/20)

Remove one attack class from all training

Natural novelty

Temporal ordering

New classes first seen after training

4 rolling cut-offs (30 / 45 / 60 / 75%)

Purged-block fallback (short captures)

10-min blocks → **60 / 20 / 20** (train / val / test) → Purge adjacent blocks → Min. test windows

3. DETECTOR PANEL & MULTI-SEED EVALUATION

12 detectors / 5 families

- **Confidence**
  MSP, Energy, P(attack)
- **Representation**
  Stress, Mahalanobis, kNN
- **Benign-only**
  IF, OCSVM, PCA, AE
- **Contrastive**
  CLAD
- **Reference**
  XGBoost, XGBoost-s

Recurrent backbones
(confidence and representation)

- AHSD
- GRU

Multi-seed evaluation

5 seeds → **210** runs → **1** code commit → **Pre-** registered

hypotheses, metrics, gates

4. EVIDENCE & ACCEPTANCE

Metrics

AUC | TPR at 1% FPR | PR-AUC | Seed s.d. | Family ranking | Checkpoint sensitivity

Three main findings

Protocol changes the winner
LOACO winner differs from natural novelty

High AUC, most attacks missed
AUC ≤ 0.75: ≤ 18% caught at 1% FPR

One run is unreliable
Seed variation up to 17×

HoNE acceptance protocol: 8 checks

- ✓ Leakage controls
- ✓ Natural novelty + LOACO
- ✓ ≥ 5 seeds
- ✓ Valid novelty scores
- ✓ Benign-only + XGBoost
- ✓ Operating-point metrics
- ✓ Checkpoint sensitivity
- ✓ Pre-registration / log

Operator review → **TRUST** | **REQUEST EVIDENCE**